

How to proceed: priorities, new results and research directions

Nisha Singh
Shiv Nadar Institution of Eminence

24 September 2026

In one paragraph

The three papers are sound and close to submission; what they need is a few days of cleaning, not new physics. The most important change is in how paper 1 is *framed*. My new calculations (Sec. 2) show that cooling does not really measure L_{sym} : it measures the conditions at about twice saturation density, i.e. the proton fraction $x_p(2n_0)$, the symmetry energy $S(2n_0)$, and through them the direct Urca threshold mass M_{DU} . L_{sym} is a model-dependent translation of that. Written this way, the result agrees with Montefusco et al. [1] on the physical quantity (the onset mass) and explains the L_{sym} tension instead of being weakened by it. The next research step (paper 4) is to run the same cooling analysis with an EoS family in which L_{sym} and $S(2n_0)$ are not locked together.

1 Immediate tasks (1–2 weeks)

1. **Send me** `Data/cooling_chi2_final.csv` (or run it yourself):

```
python3 combine_cooling_lab.py Data/cooling_chi2_final.csv
```

This gives the credible intervals for E_{sym} , L_{sym} , $S(2n_0)$, $x_p(2n_0)$, $R_{1.4}$ and M_{DU} , the posterior probability $P(M_{\text{DU}} \leq 1.4 M_{\odot})$, and the same with the INDRA–FAZIA constraint added (Sec. 3). M_{DU} is currently known for 683 of the 864 EoS; run the script on the Mac, where all `MRcurve_*_CAT.dat` files are, after regenerating `reports/data/eos_physical_quantities.dat`, or send me the remaining curves.

2. **Paper 2:** wait for the neutron-skin rerun (about 10 hours), then regenerate all its numbers (I will do this).
3. **Fix the citations** marked % CHECK in the drafts: the RX J0822 sources (Table 1 of Potekhin et al. 2020), the published ζ values.
4. **Quick robustness runs** (no new NSCool runs needed): RX J0822 with $\sigma = 0.12$ dex; the 3P_2 -gap alternative (`durca_sf_altB_vc`, already run) as a second model in a Bayesian average instead of a separate check.
5. **Confirm the Landau mass** in the tables actually used (two `grep` commands in the analysis note).
6. **Write to D. Page** about the `bf_r` problem before paper 3 is submitted, and ask which NSCool versions are affected.

2 What cooling actually measures

For all 864 EoS I computed $S(2n_0)$, $S(3n_0)$, $x_p(2n_0)$, $x_p(3n_0)$ and K_{sym} with the EoS code, and M_{DU} from the pipeline M – R curves (`reports/data/eos_physical_quantities.dat`). The correlation of the threshold mass with each quantity:

	L_{sym}	E_{sym}	$S(2n_0)$	$S(3n_0)$	$x_p(2n_0)$	$x_p(3n_0)$
correlation with M_{DU}	−0.69	−0.72	−0.82	−0.71	−0.85	−0.76

At the *same* $E_{\text{sym}} = 32.5$ and $L_{\text{sym}} = 65$ MeV the five models give:

Model	$S(2n_0)$ (MeV)	$x_p(2n_0)$	M_{DU} ($M_{\odot}$)
IOPB-I	53.0	0.114	1.54
FSUGarnet	53.2	0.113	1.57
BigApple	54.1	0.115	2.05
GM1	49.1	0.099	no DUrca
GM2	47.2	0.093	no DUrca

And conversely, the EoS with $M_{\text{DU}} = 1.45$ – $1.55 M_{\odot}$ span $L_{\text{sym}} = 50$ – 85 MeV but only $x_p(2n_0) = 0.111$ – 0.129 . So:

- the cooling data select the proton fraction around $2n_0$ (and hence M_{DU});
- the L_{sym} interval is the image of that selection through five specific Lagrangians, and depends on them;
- this is why the L_{sym} values of cooling and of laboratory analyses can differ while the onset masses agree (1.77 vs $1.89 M_{\odot}$).

Suggestion for paper 1: quote $x_p(2n_0)$, $S(2n_0)$ and M_{DU} as the primary, less model-dependent result, and L_{sym} as the model-family translation. This makes the paper stronger, not weaker, against the comparison with [1].

3 Combining cooling with the heavy-ion constraint

Montefusco et al. publish, for use by others, a Gaussian form of their INDRA–FAZIA constraint in $(E_{\text{sym}}, L_{\text{sym}}, K_{\text{sym}})$ (their Appendix D). I applied it to our grid:

- within our model family it prefers $E_{\text{sym}} \approx 28$ – 30 MeV and $L_{\text{sym}} \approx 40$ – 55 MeV;
- 28 EoS inside the cooling 68% region are within 1σ of it (the closest is GM2 at $E_{\text{sym}} = 29.5$, $L_{\text{sym}} = 50$), and 64 within 2σ ;
- our single best fit (GM2 at 32.5/80) is *not*: its E_{sym} is $\approx 3.8\sigma$ above theirs.

So cooling and heavy-ion data are compatible, and together they point to $E_{\text{sym}} \approx 29$ MeV and $L_{\text{sym}} \approx 50$ MeV. The combination is a natural new result for paper 1 (a paragraph and a figure). It needs only the final cooling χ^2 file; the script `combine_cooling_lab.py` is ready and tested.

Note: the dry run of the script on the old (frozen-tail, one-mode) χ^2 file shows the expected behaviour: adding the heavy-ion constraint pulls E_{sym} to 28–30 MeV and makes DUrca in $\leq 1.4 M_{\odot}$ stars very unlikely. Those numbers are not results; only the final χ^2 file gives real ones.

4 The main limitation, and paper 4

Below $L_{\text{sym}} = 30$ MeV the present model family essentially stops: GM1 gives no usable EoS, GM2 survives at 20–25 MeV but DUrca then never opens, and FSUGarnet, IOPB-I and BigApple end at 35–50 MeV (no real g_ρ). Therefore the region preferred by the nuclear-structure part of [1] ($L_{\text{sym}} \approx 20$ –37 MeV) cannot be tested with our current models. And in all five models the ω - ρ term ties L_{sym} rigidly to $S(2n_0)$.

Proposal for paper 4: repeat the cooling analysis with an EoS family where L_{sym} and the high-density symmetry energy are independent. Three options, in order of effort:

1. **Density-dependent RMF** (DDME2, DDMEX, DD2 type), where the ρ coupling falls with density as $g_\rho(n) = g_\rho(n_0) \exp[-a_\rho(n/n_0 - 1)]$. a_ρ controls the high-density behaviour separately from L_{sym} . This is exactly the parameter Thapa and Sinha tuned for PREX-2 [2], so it links naturally to their work (and is a possible collaboration). The EoS code needs rearrangement terms in the chemical potentials; the rest of the pipeline is unchanged.
2. **Adding the scalar-isovector δ meson.** It changes the high-density symmetry energy and splits the proton and neutron Dirac masses, which enters the Landau masses and hence the cooling rates. Recent work already studies its effect on DUrca.
3. **A metamodel-type EoS** as in [1], with L_{sym} and K_{sym} , Q_{sym} free. This would allow a like-for-like comparison with their result, with cooling as the new ingredient.

The key question for paper 4: *with L_{sym} and $S(2n_0)$ decoupled, does cooling still prefer $L_{\text{sym}} \approx 50$ –75 MeV, or does it only constrain $x_p(2n_0)$?* Either answer is publishable, and the second would reconcile cooling with the laboratory results.

5 Further improvements of the cooling analysis

1. **Finer mass grid.** Cooling is run at 1.0, 1.4, 1.8, 2.0 $M_\odot$ only, while the TOV profiles already exist every 0.1 $M_\odot$. The jump from 1.4 to 1.8 $M_\odot$ is exactly where DUrca switches on (median $M_{\text{DU}} = 1.77 M_\odot$), so the star-by-star mass choice is coarse there. Running 1.1–1.9 $M_\odot$ in steps of 0.1 for the final mode costs about $864 \times 5 \times$ a few seconds per run, i.e. a few hours on four workers, and allows a continuous mass (with the $1.4 \pm 0.3 M_\odot$ prior as a proper Bayesian prior).
2. **Treat each star’s mass as a nuisance parameter** and marginalise (sum over masses with the prior) instead of taking the minimum. This is the standard Bayesian choice and removes the discreteness.
3. **Age uncertainties.** Characteristic spin-down ages can be wrong by factors of a few; kinematic and remnant ages are better. Add an age error per star (larger for characteristic ages) through the local slope of the cooling curve.
4. **Average over pairing models** (gap “a” and “b”, which are both already run with heating) with equal prior weight, instead of fixing one.
5. **Use more information on Cas A.** Besides its temperature, the observed temperature change over two decades has been used to constrain superfluidity (e.g. [3, 4]). Its current status is debated because of instrumental effects, so it should enter as a systematic check, not a main constraint.
6. **Report the fit quality honestly per star** (a table of χ^2_i at the best fit): it shows at once which stars drive the result.

6 Suggested order of publication

1. **Paper 1 and paper 3 together** (same data set and pipeline; paper 3 is the robustness check of paper 1 and also reports a code problem that others need to know about). Paper 1 with the reframing of Sec. 2 and the heavy-ion combination of Sec. 3.
2. **Paper 2** after the skin rerun. Its message (the models give nearly equal Pb and Ca skins; cooling does not choose between PREX-2 and CREX) is robust.
3. **Paper 4**: density-dependent or δ -meson family through the same pipeline.

References

- [1] G. Montefusco et al., *Ruling out nucleonic direct Urca cooling in low-mass neutron stars using nuclear data*, [arXiv:2609.24940](#) (2026).
- [2] V. B. Thapa and M. Sinha, *Direct URCA process in light of PREX-2*, [arXiv:2203.02272](#).
- [3] P. S. Shternin, D. G. Yakovlev, C. O. Heinke, W. C. G. Ho and D. J. Patnaude, *Mon. Not. R. Astron. Soc.* **412**, L108 (2011), [doi:10.1111/j.1745-3933.2011.01015.x](#).
- [4] W. C. G. Ho, K. G. Elshamouty, C. O. Heinke and A. Y. Potekhin, *Phys. Rev. C* **91**, 015806 (2015), [doi:10.1103/PhysRevC.91.015806](#).